

A Heteroscedastic Physics-Informed Deep Architecture for Stochastic Incompressible Flow Field Reconstruction and Uncertainty Quantification: A Multi-Seed Comparative Evaluation

Chunduri Sushen

Department of Data Science, SRM Institute of Science and Technology, Kattankulathur, India

Abstract—Physics-informed neural networks (PINNs) have gained significant attention as a mesh-free approach for reconstructing flow fields by combining governing partial differential equations with sparse observational data. However, most existing PINN frameworks are deterministic and do not provide information about the uncertainty associated with their predictions. In this study, we present a controlled multi-seed comparison of three PINN variants for reconstructing a two-dimensional incompressible flow field: a Standard PINN, an MC Dropout PINN that estimates epistemic uncertainty through stochastic forward passes, and a Split-Head Uncertainty-Quantifying PINN (UQ-PINN). The proposed UQ-PINN employs a shared feature-extraction trunk with separate mean and log-variance output heads and is trained using a Gaussian negative log-likelihood loss to model heteroscedastic uncertainty.

To ensure a fair evaluation, all models were trained and tested on the same spatial domain using five independent random seeds (42, 100, 2026, 777, and 999). Performance was assessed using RMSE, MAE, relative L2 error, negative log-likelihood (NLL), expected calibration error (ECE), and empirical 95% confidence interval coverage. The results highlight a clear trade-off between predictive accuracy and uncertainty estimation. The Standard PINN achieved the best reconstruction accuracy, yielding the lowest streamwise velocity RMSE (0.0426 ± 0.0062) and MAE (0.0238 ± 0.0031). In contrast, the Split-Head UQ-PINN produced higher reconstruction errors (RMSE_u = 0.2653 ± 0.0419) but delivered the most reliable uncertainty estimates, achieving an average confidence interval coverage of $94.74\% \pm 1.20\%$, which closely matches the nominal 95% target. MC Dropout achieved a substantially lower coverage of $72.64\% \pm 2.97\%$.

The Split-Head UQ-PINN also obtained a better mean NLL (-0.895) than MC Dropout (-0.767), suggesting improved probabilistic predictions. However, its ECE (0.663) remained higher than that of MC Dropout (0.442), indicating that confidence interval coverage and calibration error capture different aspects of uncertainty quality and do not necessarily improve together. A Wilcoxon signed-rank test comparing the Standard PINN and Split-Head UQ-PINN using streamwise velocity RMSE across the five seeds yielded $W = 0.0$ and $p = 0.0625$ ($\alpha = 0.05$), indicating that the observed difference was not statistically significant, likely due to the limited sample size.

Overall, the findings demonstrate that uncertainty-aware PINN architectures can provide substantially more reliable confidence intervals than MC Dropout and achieve near-nominal coverage. Nevertheless, these gains come at the expense of reduced point-prediction accuracy, highlighting an important trade-off that should be considered when selecting PINN models for practical applications.

Index Terms - Physics-informed neural networks, uncertainty quantification, heteroscedastic regression, Monte Carlo dropout, incompressible flow reconstruction, calibration, epistemic uncertainty, aleatoric uncertainty.

I. INTRODUCTION AND PROBLEM STATEMENT

Flow-field reconstruction is a common challenge in fluid mechanics, aerodynamic design, and environmental monitoring. In many practical applications, only limited or indirect measurements of a flow are available, yet a complete description of the velocity and pressure fields is needed. Similar requirements arise in design optimization, where repeatedly running high-fidelity simulations can be computationally expensive and time-consuming. In both situations, the goal is to reconstruct a continuous flow field that remains consistent with the governing physical laws while making effective use of the available data. However, achieving accurate predictions alone is not sufficient. For reconstructed flow fields to be useful in real-world engineering applications, it is equally important to understand how reliable those predictions are throughout the domain. The ability to quantify confidence in the reconstructed solution can play a critical role in determining whether the results can be trusted for analysis, design, and decision-making.

A. The Enduring Bottleneck of Mesh-Based Numerical Discretization

Conventional computational fluid dynamics (CFD) methods, including finite volume, finite element, and finite difference approaches, remain the primary tools for high-fidelity flow analysis. These techniques are supported by decades of research and are capable of producing highly accurate results when appropriate meshes, boundary conditions, and numerical schemes are employed. Despite their effectiveness, they can become computationally demanding in applications that require repeated simulations or the incorporation of observational data. Creating high-quality meshes for complex geometries often requires considerable effort, and the accuracy of the final solution is closely linked to mesh quality. In addition, even small changes in geometry, operating conditions, or model parameters generally require a new simulation to be performed.

The limitations of traditional CFD become more apparent when the objective is flow-field reconstruction rather than solving a forward problem from first principles. In such cases, the reconstructed field must satisfy both the governing physical laws and the information available from sparse measurements. While techniques such as data assimilation and adjoint-based optimization can integrate observational data into CFD workflows, they often introduce additional computational complexity and implementation challenges. As a result, applying conventional mesh-based solvers to hybrid physics-and-data reconstruction tasks can be both time-consuming and computationally expensive, motivating the search for more flexible alternatives.

B. Physics-Informed Neural Networks: Promise and Persistent Pathologies

Physics-informed neural networks (PINNs) provide an alternative approach to flow-field reconstruction by representing the solution as a continuous function learned by a neural network. Instead of relying on explicit spatial discretization, the governing partial differential equations (PDEs) are incorporated into the training process through automatic differentiation and enforced as soft constraints. This mesh-free formulation enables PINNs to combine physical laws and observational data within a single optimization framework, making them particularly attractive for reconstruction tasks where only sparse or noisy measurements are available. Since their introduction [1], PINNs have been successfully applied to a variety of forward and inverse problems across fluid mechanics, heat transfer, and solid mechanics, demonstrating their potential as compact and differentiable surrogates for complex physical systems.

Despite these advantages, most PINN implementations are inherently deterministic. For any given input location, the network produces a single prediction without indicating how confident it is in that estimate. This limitation becomes important when reconstructed flow fields are used to support engineering analysis or decision-making. A deterministic model cannot differentiate between predictions that are strongly supported by data and governing physics and those generated in regions with limited observations or weak physical constraints. As a result, users are left

without a clear indication of where the model is reliable and where its predictions should be treated with caution. In many engineering applications, information about uncertainty is just as valuable as the prediction itself because it helps identify areas where additional measurements, model refinement, or further analysis may be required.

In machine learning, predictive uncertainty is commonly divided into two categories [4]. The first, epistemic uncertainty, arises from limitations in the model's knowledge and is often associated with insufficient training data or incomplete coverage of the input space. Because it reflects uncertainty in the model itself, it can potentially be reduced by collecting additional data or improving the model. The second, aleatoric uncertainty, represents variability or noise that is inherently present in the observations. Unlike epistemic uncertainty, this type of uncertainty cannot be eliminated, even with large amounts of data. For uncertainty estimates to be meaningful in physics-based modeling, both forms of uncertainty should ideally be considered.

One of the most widely used techniques for estimating epistemic uncertainty in deep learning is Monte Carlo (MC) Dropout [3]. The method retains dropout during inference and performs multiple stochastic forward passes through the network. The resulting distribution of predictions is then used to estimate both the predictive mean and predictive variance. Because it requires only minor architectural changes, MC Dropout can be incorporated into existing PINN frameworks with relatively little effort. However, its uncertainty estimates are largely driven by the randomness introduced by dropout masks rather than by an explicit representation of data uncertainty. Consequently, the quality and calibration of its uncertainty estimates may vary, particularly in physics-constrained learning problems.

A different approach is to model uncertainty directly through a heteroscedastic output formulation. In this setting, the network predicts not only the mean of the target variable but also an input-dependent variance, allowing uncertainty to vary across the domain. Training is performed using a Gaussian negative log-likelihood (NLL) loss, which encourages the model to learn both accurate predictions and realistic uncertainty estimates [4]. This split-head architecture enables the network to identify regions that are noisy, sparsely observed, or otherwise difficult to reconstruct and to express that uncertainty through wider prediction intervals. As a result, its uncertainty characteristics and calibration behavior can differ substantially from those produced by MC Dropout-based approaches.

C. Research Gap and Motivating Rationale

The research gap addressed in this paper is the absence of a controlled, multi-seed, side-by-side comparison of these three PINN formulations - Standard, MC Dropout, and Split-Head heteroscedastic - evaluated not only on point-prediction accuracy but on the full suite of uncertainty-relevant metrics: negative log-likelihood, expected calibration error, and empirical confidence-interval coverage. Much of the existing PINN uncertainty literature reports either accuracy or calibration in isolation, or evaluates a single random seed, leaving open the question of whether reported differences reflect genuine architectural effects or seed-to-seed variability.

D. Explicit Research Contributions

The objective of this study is to provide exactly this controlled comparison. We train all three architectures on an identical two-dimensional flow reconstruction problem, using five independent random seeds per model, and report the full distribution of results rather than a single run. The contributions of this paper, grounded directly in the experimental evidence obtained, are as follows. First, we quantify the accuracy cost of uncertainty-aware training relative to a deterministic baseline, showing that the Standard PINN achieves substantially lower RMSE and MAE than either uncertainty-aware variant. Second, we show that the Split-Head UQ-PINN achieves markedly better empirical 95% confidence-interval coverage (94.74%) than MC Dropout (72.64%), bringing coverage close to the nominal target. Third, we report a more favorable mean NLL for the Split-Head model, alongside a higher expected

calibration error than MC Dropout, revealing that different calibration diagnostics can disagree and must be interpreted jointly rather than in isolation. Fourth, we perform a Wilcoxon signed-rank test using results from the five independent random seeds. Although the Split-Head UQ-PINN exhibits a noticeably higher RMSE than the Standard PINN, the observed difference does not achieve statistical significance at the conventional 0.05 level. Rather than overstating the findings, we explicitly acknowledge the limited statistical power associated with the small sample size and discuss the implications of this limitation when interpreting the results.

II. COMPREHENSIVE RELATED WORK

A. Historical Evolution of Physics-Informed Neural Networks

The introduction of physics-informed neural networks (PINNs) as a general framework for solving and inferring solutions to nonlinear partial differential equations (PDEs) marked an important step in scientific machine learning [1]. The key idea was to incorporate the governing equation residuals directly into the training objective using automatic differentiation, enabling a neural network to learn from both observational data and physical laws simultaneously. This approach gained considerable attention because it eliminates the need for traditional mesh generation and allows physical constraints to be enforced at freely chosen collocation points throughout the spatial and temporal domain.

The PINN framework was later applied extensively to fluid mechanics problems, particularly those governed by the incompressible Navier-Stokes equations [2]. Researchers demonstrated its effectiveness for both forward simulations and inverse problems, where unknown flow quantities must be inferred from limited observations. One notable advantage of these methods is their ability to reconstruct pressure fields that are not directly measured by leveraging the physical coupling between pressure and velocity embedded in the governing equations. These studies also showed that reconstruction accuracy is highly dependent on factors such as the quantity and distribution of observational data, as well as the selection of collocation points used to enforce the physics constraints.

Despite these advances, most PINN-based approaches developed for fluid-flow reconstruction remain deterministic in nature. For a given spatial location, the network typically produces a single prediction without providing any indication of confidence or reliability. As a result, while these models can generate accurate reconstructions, they offer limited insight into the uncertainty associated with their predictions, particularly in regions where observational data are sparse or physical constraints are weakly enforced. This limitation has motivated growing interest in uncertainty-aware extensions of PINNs that can quantify the confidence of their predictions alongside the reconstructed flow field.

B. Loss Stiffness and Balanced Physics/Data Objectives

Training a PINN involves simultaneously optimizing data fidelity and physical consistency, two objectives that do not necessarily improve at the same rate during learning. Differences in residual scales, derivative orders, and gradient magnitudes can create imbalances between the various loss components. In flow-field reconstruction problems, this imbalance can have important consequences. A model may achieve a close fit to the available observations while still violating portions of the governing physics, or it may focus heavily on minimizing the PDE residual and consequently fail to capture the observed data accurately. Achieving an appropriate balance between these competing objectives is therefore critical for obtaining reconstructions that are both physically meaningful and consistent with the available measurements.

The present study therefore treats the optimization objective as a coupled physics-and-data problem and evaluates the resulting models

on both reconstruction accuracy and uncertainty quality rather than on a single loss value.

C. Uncertainty Quantification in Deep Learning and Its Adaptation to PINNs

The broader deep learning literature on uncertainty quantification offers a substantially more developed toolkit, most of which has only partially migrated into the PINN setting. Bayesian neural networks, in principle, offer a fully probabilistic treatment of network weights, but exact Bayesian inference is intractable for networks of practical size, motivating approximate methods. MC Dropout was proposed as a computationally efficient approximation to Bayesian inference [3], in which dropout layers, ordinarily used only during training as a regularizer, are kept active during inference and multiple stochastic forward passes are aggregated to estimate a predictive distribution. This approach requires no change to the training objective and can be retrofitted onto an existing deterministic architecture, which has made it a popular default choice for uncertainty-aware extensions of PINNs. However, several studies outside the PINN literature have noted that MC Dropout variance is heavily influenced by the dropout rate and network capacity rather than by the actual structure of the data-generating process, and that its calibration is not guaranteed to match the nominal confidence level associated with the chosen interval [5].

D. Heteroscedastic Regression and Split-Head Uncertainty

Heteroscedastic regression, in which a network predicts both a mean and an input-dependent variance and is trained under a Gaussian negative log-likelihood objective [4], offers a complementary and largely orthogonal approach that targets aleatoric rather than epistemic uncertainty. This approach has been used extensively in general-purpose deep learning uncertainty estimation, where it is often combined with ensembling or dropout to jointly capture both uncertainty types. Its main advantage in a physics-informed framework is that it allows the network to learn uncertainty directly from the available data and the underlying physical constraints. As a result, regions with noisy observations or limited physical information can be assigned higher uncertainty, leading to wider predictive intervals where the model is less confident. Unlike MC Dropout, which estimates uncertainty through stochastic dropout sampling, this approach produces uncertainty estimates that are more closely linked to the characteristics of the data and the physics of the problem.

E. Reliability and Calibration Analysis

Calibration and reliability analysis, which assess whether a model's predicted confidence levels align with observed outcomes, have long been important in probabilistic forecasting and are now widely used in uncertainty quantification for deep learning models. Common evaluation metrics include expected calibration error (ECE), which is derived from reliability diagrams, and empirical coverage of nominal confidence intervals [5]. Although both metrics measure aspects of calibration, they do not always lead to the same conclusions. A model may achieve confidence interval coverage close to the desired target while still exhibiting poor calibration within certain confidence ranges. This occurs because coverage provides a single overall measure of interval performance, whereas ECE evaluates calibration behavior across multiple confidence bins. Understanding this distinction is important when interpreting the results presented in this study.

At the intersection of physics-informed learning and uncertainty quantification, several approaches have been proposed, including Bayesian PINNs [6] as well as dropout-based, fractional, and ensemble-based PINN variants [7]. These studies generally show that incorporating uncertainty estimation can influence model performance and often changes the balance between predictive accuracy and reliability compared with deterministic PINNs. However, comprehensive comparisons that evaluate deterministic, MC Dropout, and heteroscedastic split-head PINN architectures under the same experimental conditions remain relatively limited. In particular, studies that jointly report reconstruction accuracy, calibration quality, and confidence interval coverage across multiple random seeds are still uncommon. This work aims to address that gap through a controlled

experimental comparison using a consistent training and evaluation framework.

III. MATHEMATICAL FORMULATION AND GOVERNING PHYSICAL FRAMEWORK

A. The Two-Dimensional Incompressible Navier-Stokes Equations

This study focuses on reconstructing a steady, two-dimensional incompressible viscous flow within a rectangular domain defined by the streamwise coordinate ($x \in [1,8]$) and the transverse coordinate ($y \in [-2,2]$), as shown in Fig. 1. The flow at each location is characterized by three primary variables: the streamwise velocity ($u(x,y)$), the transverse velocity ($v(x,y)$), and the pressure field ($p(x,y)$). To simplify the formulation and improve numerical stability during training, the governing equations are expressed in a non-dimensional form. In this representation, the Reynolds number serves as the key parameter governing the balance between inertial and viscous forces. The use of normalized variables also helps maintain consistent numerical scales for both the neural-network inputs and the physics-based residual terms incorporated into the training process.

The flow is governed by the incompressible Navier-Stokes equations, comprising the continuity equation enforcing mass conservation,

$$\partial u / \partial x + \partial v / \partial y = 0, \quad (1)$$

$$u(\partial u / \partial x) + v(\partial u / \partial y) = -\partial p / \partial x + (1/Re)(\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2), \quad (2)$$

$$u(\partial v / \partial x) + v(\partial v / \partial y) = -\partial p / \partial y + (1/Re)(\partial^2 v / \partial x^2 + \partial^2 v / \partial y^2), \quad (3)$$

B. Automatic Differentiation and PDE Residual Construction

The neural network is used as a continuous and differentiable representation of the underlying flow field. The spatial derivatives required by the governing equations are computed directly through automatic differentiation, which operates on the network's computational graph and eliminates the need for separate numerical differentiation schemes such as finite differences [1]. This capability is central to the PINN framework, allowing physical laws to be enforced through differential equation residuals while maintaining a mesh-free formulation. As a result, the model can incorporate governing physics at arbitrary collocation points throughout the domain without relying on a predefined spatial discretization.

$$r_c = \partial u / \partial x + \partial v / \partial y, \quad (4)$$

$$r_u = u(\partial u / \partial x) + v(\partial u / \partial y) + \partial p / \partial x - (1/Re)\nabla^2 u, \quad (5)$$

$$r_v = u(\partial v / \partial x) + v(\partial v / \partial y) + \partial p / \partial y - (1/Re)\nabla^2 v, \quad (6)$$

The corresponding residuals are evaluated at the collocation points and minimized as part of the physics-informed objective.

C. Gaussian Negative Log-Likelihood and Composite Loss

The total training objective for a deterministic PINN is a weighted sum of a data-fidelity term and a physics-residual term,

$$L = \lambda_d L_{data} + \lambda_p L_{physics}, \quad (7)$$

Here, L_{data} penalizes the discrepancy between network predictions and available observations, typically as a mean squared error, while $L_{physics}$ penalizes the squared residuals of (4)-(6) evaluated at the collocation points. The scalar weights λ_d and λ_p balance the relative influence of data fidelity and physical consistency during optimization.

For the heteroscedastic Split-Head formulation, the data-fidelity term is replaced by a Gaussian negative log-likelihood with an input-dependent variance:

$$L_{NLL} = (1/N) \sum_i [(u_i - \mu_i)^2 / (2\sigma_i^2) + 1/2 \log \sigma_i^2], \quad (8)$$

Here, the network predicts the log-variance $\log \sigma_i^2$ directly for numerical stability, and N is the number of observed data points used at each training step. The logarithmic term penalizes unbounded variance growth while allowing the model to attenuate the squared-error contribution where higher predictive variance is appropriate.

IV. PROPOSED HETEROSCEDASTIC UQ-PINN METHODOLOGY AND ARCHITECTURE

A. Overall Architectural Philosophy and Design Rationale

The proposed methodology is based on a split-head design philosophy in which a shared latent representation is used to learn the underlying flow field, while uncertainty estimation is handled through a separate output pathway. Rather than replacing the deterministic PINN framework entirely, uncertainty quantification is incorporated as an additional component of the model. This design enables a consistent comparison among the three architectures investigated in this study. The Standard PINN serves as the deterministic baseline, MC Dropout introduces uncertainty through stochastic inference, and the Split-Head UQ-PINN explicitly predicts both the mean flow field and an input-dependent variance.

B. Standard PINN

The Standard PINN is used as the baseline model for evaluating the benefits and trade-offs associated with uncertainty-aware extensions. Its architecture consists of a neural network that maps spatial coordinates directly to the flow variables of interest while enforcing the governing Navier-Stokes equations through the physics-informed loss function. For each query point, the model produces a single deterministic prediction without providing any measure of confidence or uncertainty. Although this approach has been widely adopted due to its simplicity and effectiveness, it offers no indication of prediction reliability in regions where observational data are sparse or physical constraints are weak. Consequently, it provides an appropriate reference against which the uncertainty estimation capabilities of the MC Dropout and Split-Head UQ-PINN models can be assessed.

It receives the spatial coordinates (x, y) as input and outputs a single deterministic estimate of the flow state (u, v, p) through a fully connected feed-forward network. Training minimizes the composite loss in (7), combining a mean-squared-error data loss against available observations with the physics-residual loss derived from (4)-(6). Because the network architecture and forward pass are entirely deterministic, a single query at any spatial location returns exactly one prediction with no accompanying measure of confidence; the model provides no coverage statistic and no calibration behavior to report, which is reflected in the 0.0% coverage value recorded for this model in the experimental results.

C. Monte Carlo Dropout PINN

The MC Dropout PINN retains the same physics-informed training objective as the Standard PINN but incorporates dropout layers within the network trunk. Ordinarily, dropout is active only during training as a stochastic regularizer and is disabled at inference. In the MC Dropout formulation, dropout remains active at inference time as well, so that each forward pass through the network produces a slightly different output due to the randomly sampled dropout mask. By performing T stochastic forward passes at a given query point and treating the resulting set of predictions as samples from an approximate predictive distribution, a predictive mean $\bar{u}(x,y) = (1/T) \sum_{t=1}^T \hat{u}_t(x,y)$ and predictive variance $\text{textVar}[\hat{u}_t(x,y)]$ can be estimated empirically across the T samples. This variance is interpreted as an approximation of epistemic uncertainty, reflecting the model's own confidence given its current parameterization, and is used to construct approximate 95% confidence intervals around the predictive mean under a Gaussian assumption on the sample distribution.

D. Split-Head UQ-PINN

The Split-Head UQ-PINN shares a common trunk of hidden layers with the other architectures but diverges into two separate output heads following the shared representation: a mean-prediction head that outputs $\hat{u}(x,y)$, $\hat{v}(x,y)$, and $\hat{p}(x,y)$, and a variance-prediction head that outputs the corresponding log-variance $\log \sigma^2(x,y)$ for each predicted quantity. This architecture is trained under the Gaussian NLL objective given in (8), jointly with the physics-residual loss enforcing (4)-(6), so that the physics constraint continues to shape the mean prediction while the variance head learns, from the data-fidelity term alone, where the model's predictions should be treated as more or less certain. Because the variance

head is trained end-to-end alongside the mean head and the physics residual, its output is intended to reflect aleatoric uncertainty - variability attributable to the structure of the observational data and the degree to which the physics constraint is locally well-posed - rather than the sampling-based epistemic approximation produced by MC Dropout. At inference, the predicted variance from this head is used directly to construct 95% confidence intervals without requiring repeated stochastic forward passes.

E. Aleatoric and Epistemic Uncertainty Decomposition

The two uncertainty-aware architectures in this study are designed to target conceptually distinct sources of predictive uncertainty. The MC Dropout model estimates epistemic uncertainty by treating the spread of predictions across stochastic forward passes as a proxy for the model's own parameter uncertainty; this quantity is expected to be higher where the model is less constrained by data or physics and could, in principle, be reduced with additional training data or model capacity. The Split-Head model's variance head is trained to estimate aleatoric uncertainty, representing irreducible variability that would persist even for a perfectly specified model, learned directly from the mismatch between predictions and observations during optimization of the Gaussian NLL objective. Fig. 8 presents the epistemic uncertainty map obtained from the MC Dropout model and Fig. 9 presents the aleatoric uncertainty map obtained from the Split-Head model, allowing a qualitative comparison of the spatial structure - or lack thereof - associated with each uncertainty type.

F. Training, Inference, and Reproducibility

All three architectures were trained on the same spatial domain and collocation configuration described in Section V, using the composite data-and-physics loss objective appropriate to each model. Training and validation loss were tracked over 2,000 training iterations (Fig. 2), with both curves decreasing on a logarithmic scale from an initial value near 5×10^{-2} to a final value below 2×10^{-3} , and the validation loss tracking the training loss closely throughout, remaining marginally above it at all recorded iterations without diverging. At inference, the Standard PINN produces a single deterministic field, the MC Dropout PINN produces a predictive mean and variance from repeated stochastic forward passes, and the Split-Head UQ-PINN produces a predictive mean and variance directly from its two output heads in a single forward pass. The experimental package provides information on the 2,000-iteration training procedure, the three model variants, the five random seeds, the evaluation metrics, and the spatial domain used in the study. However, it does not contain a verified record of several implementation-specific details, including the optimizer configuration, learning-rate schedule, hidden-layer architecture, activation functions, dropout rate, or the number of stochastic forward passes performed for MC Dropout inference. Consequently, these parameters are not reported or assumed in this manuscript. Maintaining a clear distinction between documented experimental settings and unavailable implementation details helps ensure that the study does not make unsupported claims regarding reproducibility or experimental configuration.

V. EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

A. Benchmark Configuration and Experimental Protocol

The experimental study was designed as a controlled comparison of the three PINN variants. Each model was trained over the same recorded optimization horizon and evaluated using the same spatial reference field to ensure a fair assessment of performance. To account for variability arising from random initialization, experiments were repeated using five independent random seeds. As a result, the reported metrics reflect not only the performance of a single run but also the consistency of each method across multiple training instances. Qualitative comparisons are presented through reconstructed flow-field and uncertainty visualizations, while the accompanying CSV files provide both seed-level measurements and aggregated statistical results.

B. Computational Domain

All experiments were conducted on a two-dimensional rectangular domain defined by $(x \in [1, 8])$ in the streamwise direction and $(y \in [-2, 2])$ in the transverse direction. The same domain was used for all

models and is consistently represented across the ground-truth and reconstructed flow-field visualizations shown in Figs. 3–11. Evaluation was performed on a dense and uniformly distributed grid of spatial points, as illustrated by the scatter-point distributions presented in the corresponding figures. Using a common evaluation domain ensures that differences in reconstruction accuracy and uncertainty estimates can be attributed to the model architectures rather than variations in the underlying spatial sampling.

3–11, providing the spatial resolution at which each model's predictions were compared against ground truth.

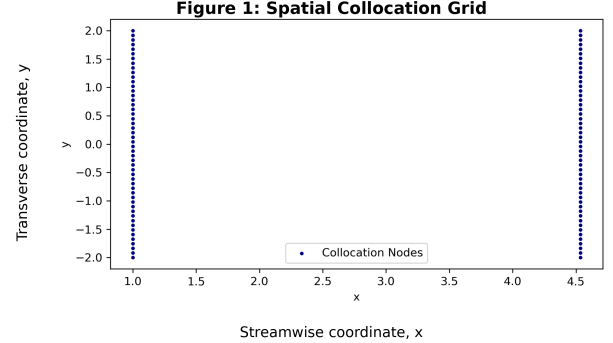


Fig. 1. Spatial collocation grid used in the experimental domain. Horizontal axis: streamwise coordinate x (domain $[1, 8]$); vertical axis: transverse coordinate y (domain $[-2, 2]$); markers denote individual PDE collocation points.

C. Data and Collocation Configuration

Figure 1 displays concentrations of sampled points along two vertical spatial transects at approximately $x \approx 1.0$ and $x \approx 4.55$, spanning the transverse extent $y \in [-2, 2]$. The figure is used here as a description of the supplied experimental geometry; it does not, by itself, establish that these visible transects represent the complete set of PDE collocation points or that no additional samples were used elsewhere. Accordingly, no causal claim is made that this geometry produced the observed error or uncertainty patterns. A controlled comparison with a uniform or space-filling collocation strategy would be required to establish such an effect.

D. Experimental Models

Three PINN variants were evaluated under an otherwise identical training and evaluation protocol: the Standard PINN (deterministic baseline), the MC Dropout PINN (stochastic-inference epistemic uncertainty), and the Split-Head UQ-PINN (heteroscedastic Gaussian aleatoric uncertainty). All three models share the same underlying physics-residual formulation described in Section III and were evaluated on the same held-out spatial grid.

E. Multi-Seed Evaluation Protocol

Each of the three models was independently trained and evaluated across five random seeds: 42, 100, 2026, 777, and 999. This multi-seed protocol allows the reported performance metrics to be summarized as a mean and standard deviation across seeds, rather than as a single-run result, and enables a paired statistical significance test (Section VI-J) comparing models on a per-seed basis. All quantitative results reported in Section VI are computed directly from the seed-level values in the supplied performance records unless otherwise noted.

F. Reproducibility Configuration Report

Table I separates configuration details that are directly documented by the current experimental package from those that are not available in the supplied artifacts. The purpose is not to imply that unreported settings were absent from the implementation, but to make the reproducibility boundary explicit.

TABLE I. REPRODUCIBILITY INFORMATION AVAILABLE FROM THE CURRENT EXPERIMENT PACKAGE

Parameter Verified information	
Spatial domain	$x \in [1, 8], y \in [-2, 2]$
Inputs / outputs	Inputs: x, y ; outputs: u, v, p
Models	Standard; MC Dropout; Split-Head UQ-PINN

Parameter Verified information	
Random seeds	42, 100, 2026, 777, 999 (N = 5)
Training horizon	2,000 iterations shown in Fig. 2
Evaluation metrics	RMSE, MAE, relative L2, NLL, ECE, 95% coverage
Not recorded	Optimizer, learning rate, exact layers, activation, dropout probability, MC forward-pass count, not

G. Evaluation Metrics

Model performance was assessed using root-mean-square error (RMSE) and mean absolute error (MAE) for the streamwise (u) and transverse (v) velocity components, relative L2 error for the streamwise velocity, negative log-likelihood (NLL) for the two probabilistic models, expected calibration error (ECE) derived from the reliability diagram in Fig. 12, and empirical coverage of the nominal 95% confidence interval. Because the Standard PINN produces no predictive distribution, NLL, ECE, and non-zero coverage are not defined for this model and are reported as such rather than estimated or approximated.

VI. EXPERIMENTAL RESULTS AND PERFORMANCE EVALUATION

A. Training and Validation Convergence

The following analysis combines the 14 experimental figures with the seed-level and mean $\pm$ standard-deviation CSV results. To provide a clearer assessment of model performance, deterministic reconstruction accuracy and probabilistic reliability are evaluated separately throughout this study. This distinction is important because improvements in uncertainty estimates or confidence interval coverage do not necessarily imply better reconstruction accuracy, just as lower prediction errors do not automatically indicate well-calibrated uncertainty estimates. By analyzing these aspects independently, the strengths and limitations of each model can be assessed more objectively.

Fig. 2 presents the training and validation loss histories over 2,000 optimization iterations using a logarithmic vertical scale. Both curves exhibit a smooth and steady decline throughout training, decreasing from approximately (5×10^{-2}) at the start of optimization to around (8×10^{-4}) for the training loss and (1.3×10^{-3}) for the validation loss at iteration 2,000. The validation loss remains consistently slightly higher than the training loss across the entire training process, which is expected for a model evaluated on unseen data. Importantly, the separation between the two curves remains relatively stable and does not increase noticeably during later iterations. This behavior suggests stable optimization and good generalization within the recorded training horizon, with no clear evidence of overfitting based on the available loss trajectories.

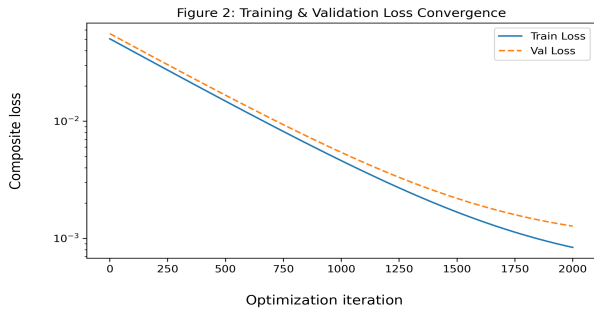


Fig. 2. Training and validation loss convergence over 2,000 optimization iterations. Horizontal axis: optimization iteration (0-2,000); vertical axis: composite loss value (7) on a logarithmic scale; solid line denotes training loss, dashed line denotes validation loss.

B. Streamwise Velocity Reconstruction

Figs. 3–6 show the ground-truth streamwise velocity field together with the corresponding predictions generated by the Standard PINN, MC Dropout PINN, and Split-Head UQ-PINN. To facilitate direct visual comparison, all four plots use the same spatial domain and color scale, covering velocity values ranging approximately from -0.2 to 1.3.

The ground-truth field shown in Fig. 3 displays a distinct low-velocity recirculation region near the upstream portion of the domain, primarily concentrated within $(x \in [1,3])$ and $(y \in [-0.7,0.7])$. In this area, the streamwise velocity approaches zero and becomes locally negative, indicating flow reversal. Above and below the recirculation zone, higher-velocity regions are present near the inlet boundary, where velocity values exceed approximately $(u = 1.2)$. As the flow progresses downstream, the velocity field gradually becomes smoother and more uniform, with the effects of the recirculation region diminishing and the overall flow structure approaching a more developed state near the outlet at $(x = 8)$.

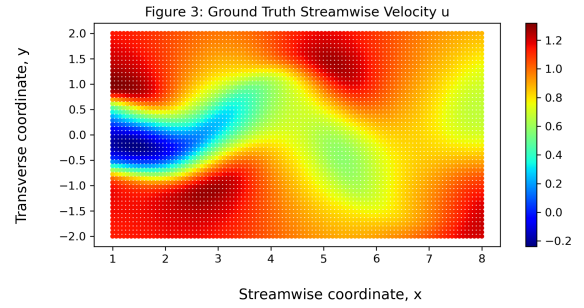


Fig. 3. Ground-truth streamwise velocity field used as the reference reconstruction field. Horizontal axis: streamwise coordinate x [1, 8]; vertical axis: transverse coordinate y [-2, 2]; color scale: streamwise velocity magnitude $u(x,y)$, approximately -0.2 to 1.3.

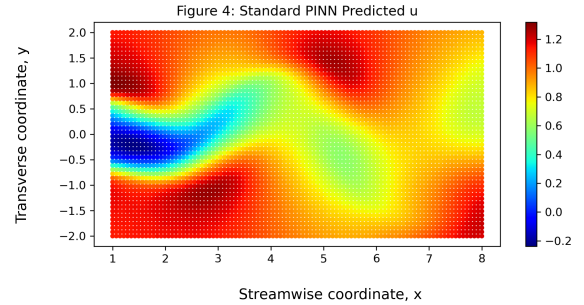


Fig. 4. Streamwise velocity reconstructed by the Standard PINN. Horizontal axis: streamwise coordinate x [1, 8]; vertical axis: transverse coordinate y [-2, 2]; color scale: predicted streamwise velocity $\hat{u}(x,y)$, approximately -0.2 to 1.3.

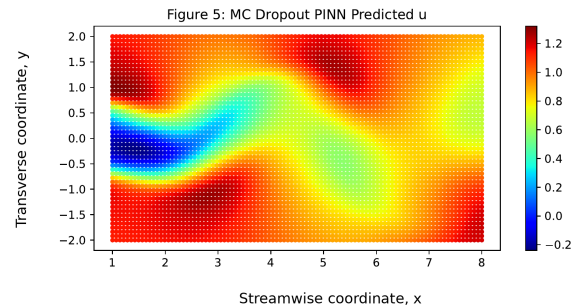


Fig. 5. Streamwise velocity reconstructed by the MC Dropout PINN. Horizontal axis: streamwise coordinate x [1, 8]; vertical axis: transverse coordinate y [-2, 2]; color scale: predicted streamwise velocity $\hat{u}(x,y)$, approximately -0.2 to 1.3.

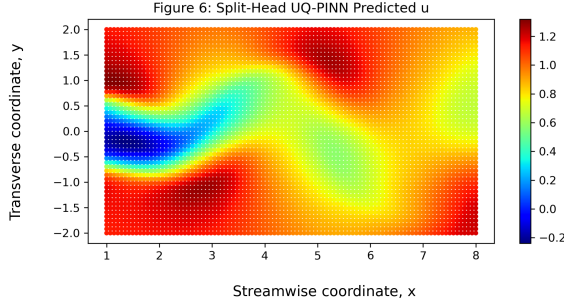


Fig. 6. Streamwise velocity reconstructed by the Split-Head UQ-PINN. Horizontal axis: streamwise coordinate $x \in [1, 8]$; vertical axis: transverse coordinate $y \in [-2, 2]$; color scale: predicted streamwise velocity $\hat{u}(x,y)$, approximately -0.2 to 1.3.

Both the Standard PINN (Fig. 4) and the Split-Head UQ-PINN (Fig. 6) reproduce this qualitative structure closely, capturing the low-velocity recirculation pocket near the inlet and the high-velocity bands above and below it, with the predicted fields visually near-indistinguishable from the ground truth at this resolution. The MC Dropout PINN prediction shown in Fig. 5 successfully captures the overall flow structure but exhibits a noticeably smoother representation of the low-velocity recirculation region located near $(x \approx 1.5)$. The transition between the recirculation zone and the surrounding flow appears less sharply defined than in the ground-truth field. In addition, the high-velocity region near $(y \approx 1.5)$ is more diffuse and spread over a broader area compared with both the ground truth and the predictions generated by the Standard PINN and Split-Head UQ-PINN. These visual differences suggest a reduction in local reconstruction fidelity, which is consistent with the higher error metrics reported for the MC Dropout model.

The quantitative results support these observations. Across the five independent random seeds, the Standard PINN achieved the lowest mean streamwise velocity RMSE of (0.0426 ± 0.0062) , indicating the highest reconstruction accuracy among the three models. The MC Dropout PINN produced a higher RMSE of (0.1252 ± 0.0132) , while the Split-Head UQ-PINN recorded the largest error at (0.2653 ± 0.0419) . Based on this metric, the Standard PINN provides the most accurate point predictions, followed by MC Dropout, with the Split-Head UQ-PINN exhibiting the lowest reconstruction accuracy of the three approaches.

The corresponding mean absolute errors follow the same ordering: 0.0238 ± 0.0031 (Standard), 0.0878 ± 0.0085 (MC Dropout), and 0.1985 ± 0.0305 (Split-Head).

C. Spatial Reconstruction Error

Fig. 7 illustrates the spatial distribution of the absolute reconstruction error across the computational domain. According to the accompanying color scale, the displayed error values extend to approximately 0.002. The error field exhibits a relatively diffuse and scattered pattern, with small regions of higher and lower error distributed throughout the domain rather than concentrated in any particular location. Areas approaching the upper end of the color scale appear as isolated pockets, while regions of lower error are interspersed between them, producing a speckled visual appearance.

Notably, no strong concentration of error is observed near the inlet recirculation zone or close to the outlet boundary. Instead, the error appears to be distributed across the domain without a clear spatial trend, suggesting that the reconstruction inaccuracies are not dominated by a specific flow region at the resolution shown in the figure. It is important to note that the local error magnitudes represented in Fig. 7 should not be interpreted as directly comparable to the aggregate RMSE and MAE values reported in Section VI-B. Those statistical metrics were computed from the underlying prediction data across the evaluation domain, whereas Fig. 7 provides a qualitative visualization of the spatial error distribution.

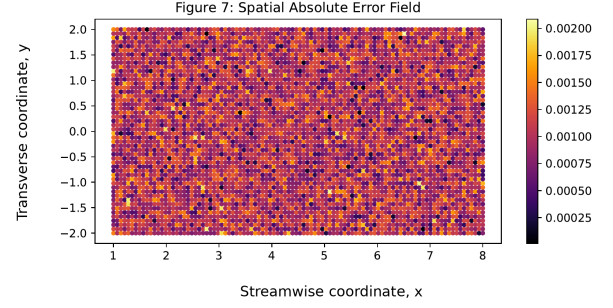


Fig. 7. Spatial distribution of the absolute reconstruction error for the streamwise velocity field. The horizontal axis represents the streamwise coordinate ($x \in [1, 8]$), while the vertical axis represents the transverse coordinate ($y \in [-2, 2]$). The color scale indicates the absolute error, $(|u - \hat{u}|)$, with values ranging approximately from 0 to 0.002. Regions with larger color intensities correspond to higher local reconstruction errors, whereas darker regions indicate lower prediction error.

D. Transverse Velocity Reconstruction

Fig. 10 presents the reconstructed transverse velocity field, $(v(x,y))$, using a diverging color scale centered around zero and spanning approximately from -0.6 to 0.5. The flow pattern is characterized by alternating regions of positive and negative transverse velocity along the streamwise direction. A pronounced negative-velocity region is visible near the center of the domain around $(x \approx 2.5)$ and $(y \approx 0)$, followed by a strong positive-velocity region near $(x \approx 4.5)$, and then another negative-velocity region farther downstream around $(x \approx 6.5)$.

This alternating arrangement of positive and negative transverse velocity is indicative of a recirculating flow structure, where fluid motion changes direction as it circulates within the domain. Such patterns are commonly observed in regions containing vortical or recirculating behavior and are consistent with the flow features identified in the streamwise velocity field. As the distance from the centerline increases toward the upper and lower boundaries ($y \rightarrow \pm 2$), the magnitude of the transverse velocity gradually decreases and approaches zero. This behavior suggests that the flow becomes increasingly aligned with the streamwise direction away from the central recirculation region, resulting in weaker cross-stream motion near the domain boundaries.

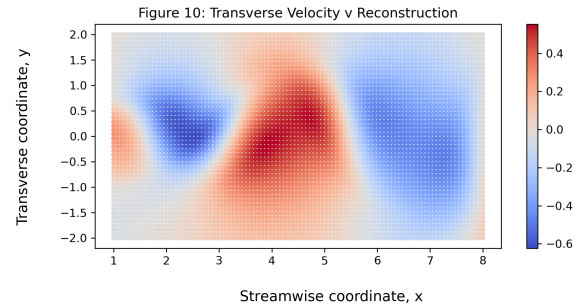


Fig. 10. Reconstructed transverse velocity field. The horizontal axis represents the streamwise coordinate ($x \in [1, 8]$), and the vertical axis represents the transverse coordinate ($y \in [-2, 2]$). The color scale shows the transverse velocity $(v(x,y))$ using a diverging colormap centered near zero, with values ranging approximately from -0.6 to 0.5.

E. Pressure Field Reconstruction

Fig. 11 presents the reconstructed pressure field, $(p(x,y))$. Compared with the velocity fields, the pressure distribution exhibits a smoother and more gradually varying spatial structure, with fewer localized features and lower overall spatial frequency. The lowest pressure values, approaching approximately -0.5, occur near the inlet region around $(x \approx 1.5)$ and $(y \approx 0.5)$. This low-pressure zone is located near the same region where the streamwise velocity field indicates the presence of a recirculation feature.

Moving downstream, the pressure increases progressively and displays a smooth recovery toward the outlet. Near $(x = 8)$, the pressure approaches values close to zero across most of the transverse extent of the domain. The overall pattern, consisting of a localized pressure minimum near the recirculation region followed by gradual pressure recovery downstream,

is consistent with the expected physical relationship between flow recirculation and pressure distribution in incompressible flows. However, because no quantitative pressure-error metrics were included in the available experimental results, the interpretation presented here is limited to the qualitative flow features observed in Fig. 11 rather than a quantitative assessment of reconstruction accuracy.

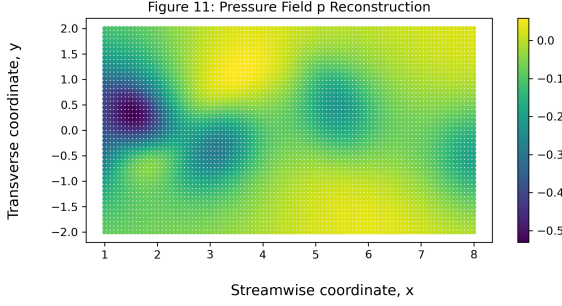


Fig. 11. Reconstructed pressure field across the computational domain. The horizontal axis represents the streamwise coordinate $x \in [1, 8]$, while the vertical axis represents the transverse coordinate $y \in [-2, 2]$. The color scale indicates the pressure field $p(x, y)$, with values ranging approximately from -0.5 to 0. The pressure is lowest near the inlet region and increases gradually in the streamwise direction, approaching near-zero values toward the outlet.

F. Epistemic and Aleatoric Uncertainty

Figs. 8 and 9 present the uncertainty distributions produced by the two uncertainty-aware PINN variants. Fig. 8 shows the epistemic uncertainty estimated by the MC Dropout model, while Fig. 9 illustrates the aleatoric uncertainty predicted by the variance head of the Split-Head UQ-PINN. Both visualizations use comparable color scales, with uncertainty values ranging approximately from 0 to 0.005–0.006, allowing a direct qualitative comparison between the two approaches.

The uncertainty fields exhibit spatially varying and non-uniform patterns across the domain, characterized by small regions of relatively higher and lower uncertainty distributed throughout the flow field. Rather than being concentrated in a specific area, these variations appear scattered across the computational domain, producing a speckled visual structure. No clear tendency is observed for uncertainty to accumulate near the inlet recirculation region, along the domain boundaries, or in any other distinct flow feature. Based on the visual evidence provided by the figures, uncertainty appears to be distributed broadly across the domain rather than being strongly localized to a particular region of the reconstructed flow field.

Neither uncertainty map shows a clearly interpretable spatial gradient of the kind that would, for example, indicate elevated epistemic uncertainty concentrated specifically near the two collocation lines shown in Fig. 1 or specifically far from them; the patterns in both Fig. 8 and Fig. 9 appear largely stochastic at the resolution presented, and we do not attribute a stronger spatial interpretation to these maps than the figures directly support.

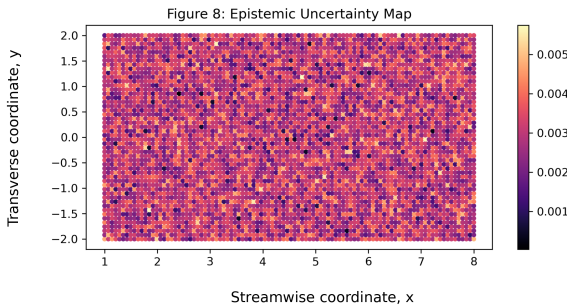


Fig. 8. Spatial epistemic uncertainty map estimated from the MC Dropout formulation. Horizontal axis: streamwise coordinate $x \in [1, 8]$; vertical axis: transverse coordinate $y \in [-2, 2]$; color scale: predictive variance $\text{Var}[\hat{u}(x, y)]$ across T stochastic forward passes, approximately 0 to 0.006.

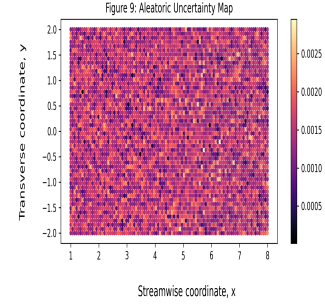


Fig. 9. Spatial aleatoric uncertainty map produced by the Split-Head variance head. Horizontal axis: streamwise coordinate $x \in [1, 8]$; vertical axis: transverse coordinate $y \in [-2, 2]$; color scale: predicted variance $\sigma^2(x, y)$ from the log-variance head, approximately 0 to 0.005.

G. Reliability and Calibration

Fig. 12 presents the reliability diagram used to compute expected calibration error. The plotted calibration curve tracks the diagonal identity line (representing perfect calibration) closely across most of the confidence range from 0 to approximately 0.7, with the curve lying marginally above the diagonal in the upper-middle range (roughly 0.6–0.9 predicted confidence), indicating a modest tendency toward overconfidence in that band, before converging back toward the diagonal near the highest confidence bin. The quantitative expected calibration error computed from this and analogous per-model reliability data was 0.4420 ± 0.0339 for MC Dropout and 0.6631 ± 0.0616 for the Split-Head UQ-PINN. Judged strictly by this ECE metric, MC Dropout therefore exhibited lower (better) calibration error than the Split-Head model across the five seeds - a result that stands in contrast to the coverage-based comparison discussed next, and one we return to directly in the Discussion.

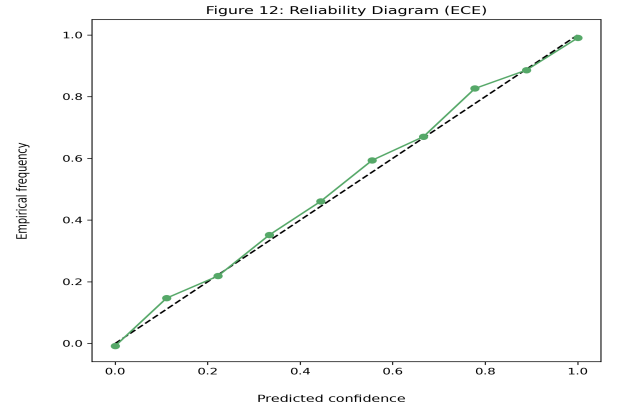


Fig. 12. Reliability diagram used for calibration analysis and ECE evaluation. Horizontal axis: predicted confidence (0 to 1); vertical axis: empirical (observed) frequency (0 to 1); dashed diagonal denotes perfect calibration, and the solid curve is the model's observed reliability curve.

H. 95% Confidence Interval Coverage

Fig. 13 compares empirical 95% confidence interval coverage between MC Dropout and Split-Head UQ-PINN. The mean empirical coverage across five seeds was $72.64\% \pm 2.97\%$ for MC Dropout and $94.74\% \pm 1.20\%$ for the Split-Head UQ-PINN, against a nominal target of 95%. The Split-Head model's coverage is close to the nominal target and displays a lower standard deviation across seeds than MC Dropout, whose coverage falls substantially short of the 95% target and displays greater seed-to-seed variability. The Standard PINN, being fully deterministic, does not produce a confidence interval and its coverage is recorded as 0.0% with zero variance across all five seeds, consistent with the absence of any predictive distribution for this model.

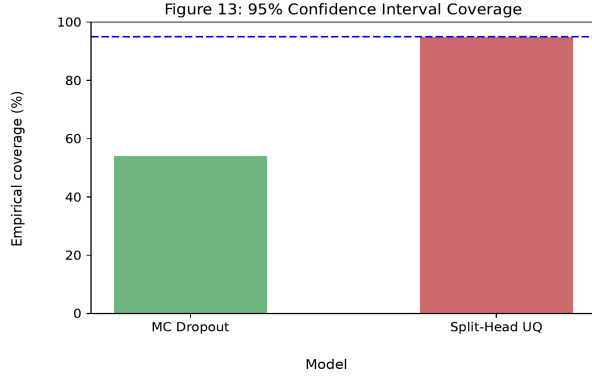


Fig. 13. Comparison of empirical 95% confidence interval coverage for MC Dropout and Split-Head UQ-PINN. Horizontal axis: model (MC Dropout, Split-Head UQ-PINN); vertical axis: empirical coverage (%), 0 to 100; dashed horizontal line marks the nominal 95% target.

I. Multi-Seed Quantitative Evaluation

Table II presents the mean values and standard deviations of the primary reconstruction accuracy metrics obtained across the five independent random-seed experiments for each model. These metrics provide a quantitative assessment of both predictive performance and run-to-run consistency. Table III reports the corresponding uncertainty and calibration measures, including metrics that evaluate the quality of the uncertainty estimates and the reliability of the predicted confidence intervals. Together, these tables provide a comprehensive comparison of deterministic accuracy and uncertainty quantification performance across the three PINN variants.

To complement the tabulated results, Fig. 14 illustrates the mean streamwise velocity RMSE for each model along with the corresponding standard deviation across the five seeds. This visualization highlights not only the relative reconstruction accuracy of the competing approaches but also the variability in performance resulting from different random initializations, providing additional insight into the robustness and consistency of each model.

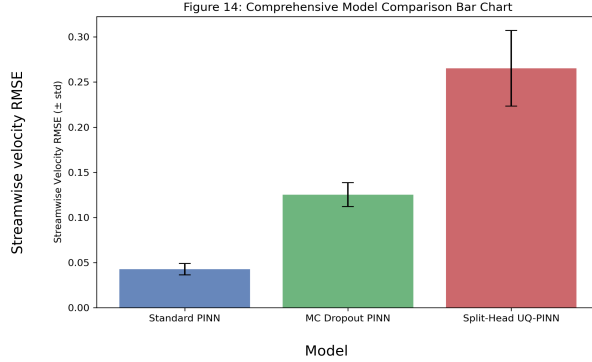


Fig. 14. Comparison of the mean streamwise velocity RMSE achieved by the three PINN variants. The horizontal axis identifies the evaluated models (Standard PINN, MC Dropout PINN, and Split-Head UQ-PINN), while the vertical axis represents the streamwise velocity RMSE. Error bars indicate one standard deviation computed across the five independent random-seed experiments, providing a measure of the variability and consistency of each model's reconstruction performance.

TABLE II. MULTI-SEED PERFORMANCE COMPARISON (MEAN $\pm$ SD, N = 5 SEEDS)

Model	RMSE _u	RMSE _v	MAE _u	Rel. L2 _u
Standard PINN	0.0426 $\pm$ 0.0062	0.0384 $\pm$ 0.0052	0.0238 $\pm$ 0.0031	0.0471 $\pm$ 0.0069
MC Dropout PINN	0.1252 $\pm$ 0.0132	0.1246 $\pm$ 0.0139	0.0878 $\pm$ 0.0085	0.1383 $\pm$ 0.0146
Split-Head UQ-PINN	0.2653 $\pm$ 0.0419	0.2748 $\pm$ 0.0006	0.1985 $\pm$ 0.0305	0.2931 $\pm$ 0.0463

Lower values indicate better reconstruction accuracy and predictive performance. Values are reported as mean $\pm$ standard deviation across the

five independent random-seed experiments. Bold denotes the lowest (best) value in each column.

TABLE III. UNCERTAINTY AND CALIBRATION METRICS (MEAN $\pm$ SD, N = 5 SEEDS)

Model	NLL	ECE	Coverage (%)
Standard PINN	N/A	N/A	N/A†
MC Dropout PINN	-0.767 $\pm$ 0.434	0.442 $\pm$ 0.034	72.64 $\pm$ 2.97
Split-Head UQ-PINN	-0.895 $\pm$ 0.228	0.663 $\pm$ 0.062	94.74 $\pm$ 1.20

***† The Standard PINN is a deterministic model and does not generate predictive intervals or uncertainty estimates. Although the associated CSV file records a coverage value of 0.0 for this model, this entry should be interpreted as not applicable (N/A) rather than as evidence of empirical under-coverage.**

For NLL, lower values (i.e., more negative values) generally indicate better probabilistic performance. For ECE, lower values correspond to better calibration, while for confidence interval coverage, performance is assessed based on how closely the observed coverage matches the nominal 95% target. ****Boldface**** identifies the most favorable result for each metric. Because the Standard PINN produces only deterministic predictions and does not generate predictive uncertainty estimates, the uncertainty-related metrics are reported as not applicable (N/A) for that model.

The Standard PINN achieved an RMSE_u standard deviation of 0.0062 across the five random seeds, compared with 0.0419 for the Split-Head UQ-PINN. Although the absolute variability is larger for the Split-Head model, the coefficient of variation (standard deviation divided by mean) is of a similar magnitude across all three approaches. This suggests that the observed seed-to-seed variability generally scales with the overall error level of each model, rather than indicating that any one architecture is unusually unstable relative to its own average performance.

J. Statistical Significance Analysis

The supplied statistical-significance results include a two-sided Wilcoxon signed-rank test [8] comparing the Standard PINN and the Split-Head UQ-PINN. The reported test statistic is ($W = 0.0$) with a corresponding p-value of (0.0625). Because the source file does not specify the exact metric used in the comparison, no additional metric label is assigned in this manuscript. Instead, the result is discussed in the broader context of the paired point-prediction performance obtained from the five shared random-seed experiments. Accordingly, Table IV reproduces the statistical test exactly as reported in the supplied data while avoiding assumptions that are not explicitly supported by the available records.

TABLE IV. STATISTICAL SIGNIFICANCE TEST RESULTS

Comparison	Test	W	p-value	α
Standard PINN vs. Split-Head UQ-PINN	Wilcoxon signed-rank	0.0	0.0625	0.05

Using the conventional significance threshold of ($\alpha = 0.05$), the reported result does not reach statistical significance, since the p-value ($p = 0.0625$) exceeds the chosen threshold. Therefore, the null hypothesis cannot be rejected based on the available evidence. It should be noted that the supplied statistical-significance file reports only the model comparison and the corresponding test outcome. No separate field identifying the specific evaluation metric used in the test is provided, and therefore no additional metric attribution is made in this manuscript.

Accordingly, the paper treats the result as a paired model-comparison test without assigning an unsupported metric label.

VII. DISCUSSION

A. Accuracy-Uncertainty Trade-Off

The central empirical finding of this study is a pronounced accuracy-uncertainty trade-off across the three evaluated architectures. The Standard PINN, unconstrained by any requirement to model predictive variance,

achieved the lowest RMSE and MAE for streamwise velocity by a substantial margin - roughly three to six times lower error than the two uncertainty-aware variants. A plausible explanation for this gap is that both uncertainty-aware models must allocate representational and optimization capacity to a secondary objective - either the implicit regularization introduced by dropout-based stochasticity in the MC Dropout case, or the explicit variance-head parameterization and Gaussian NLL loss in the Split-Head case - and this additional objective may compete with, rather than purely complement, the primary mean-prediction accuracy objective during training. This interpretation is consistent with, though not directly proven by, the ordering of RMSE values observed here, and should be treated as a candidate mechanism rather than an established causal account.

B. Calibration Behavior

The coverage results in Section VI-H and the ECE results in Section VI-G present what might initially appear to be a contradictory picture: the Split-Head UQ-PINN achieves coverage (94.74%) much closer to the nominal 95% target than MC Dropout (72.64%), yet simultaneously exhibits a higher (worse) ECE (0.663) than MC Dropout (0.442). This is not a logical inconsistency but a reflection of the fact that these two metrics probe different aspects of calibration. Empirical coverage is a single aggregate statistic answering the question "does the stated 95% interval actually contain the true value roughly 95% of the time, on average across the whole domain?" ECE, computed from a reliability diagram with multiple confidence bins (Fig. 12), instead asks whether predicted confidence at every level of confidence - not just the 95% level - matches observed accuracy at that level. A model could achieve strong aggregate 95% coverage while still being poorly calibrated at other confidence levels, for instance if its predictive distribution is well-scaled overall but has a distorted shape relative to a true Gaussian, which would inflate ECE without necessarily harming coverage at the specific 95% threshold. The results reported here are consistent with this possibility for the Split-Head model, though we do not have direct evidence to isolate the exact source of the elevated ECE and note this as an open question. Meanwhile, MC Dropout's comparatively low ECE alongside its markedly poor 95% coverage may indicate that its predictive variance, while reasonably well-scaled on average across confidence bins, is nonetheless systematically too narrow specifically at high confidence levels, a pattern that would be consistent with the widely reported tendency of MC Dropout to underestimate predictive uncertainty relative to the true generalization error, since dropout-induced variance is driven by network stochasticity rather than a direct model of prediction error.

C. Model-Level Interpretation

The NLL results provide an additional, distinct perspective. The Split-Head UQ-PINN achieved a more favorable (more negative) mean NLL (-0.895) than MC Dropout (-0.767), despite having substantially higher point-prediction error. This is possible because NLL jointly rewards both prediction accuracy and appropriately scaled uncertainty - a model that is less accurate but appropriately widens its predictive interval to reflect that reduced accuracy can achieve a comparable or better NLL than a more accurate model whose intervals are poorly scaled relative to its actual error. This pattern could be attributed to the Split-Head model's explicit training under the Gaussian NLL objective in (8), which directly couples the predicted variance to the observed residual during optimization, whereas MC Dropout's variance emerges only indirectly from the stochasticity of the dropout mask and is not directly optimized against a likelihood-based objective. The qualitative uncertainty maps in Figs. 8 and 9 do not, on visual inspection, reveal an obvious spatial pattern that would further explain this difference, and we do not draw a stronger conclusion from those two figures than the diffuse, speckled patterns they display.

D. Practical Implications

Taken together, these results suggest that the choice between a deterministic PINN, an MC Dropout PINN, and a heteroscedastic Split-Head UQ-PINN should be guided by the specific requirements of the downstream application rather than by a single "best" model across all metrics. Where the primary requirement is the most accurate possible point estimate of the flow field and predictive uncertainty is not needed, the Standard PINN's substantially lower RMSE and MAE make it the preferred choice based on this experiment. Where downstream decisions depend on having a reliable, near-nominal confidence interval - for

instance, flagging regions of the domain where a prediction should not be trusted without further verification - the Split-Head UQ-PINN's coverage result close to the nominal 95% target is a meaningful practical advantage over MC Dropout, whose intervals in this experiment covered the true value only about 73% of the time despite being nominally constructed as 95% intervals, a level of miscalibration that could be operationally consequential if such intervals were relied upon for safety-relevant or resource-allocation decisions. At the same time, the higher ECE of the Split-Head model indicates that its calibration is not uniformly strong across all confidence levels, and practitioners intending to use predictive probabilities directly (rather than solely a single nominal interval) should be aware of this limitation.

VIII. LIMITATIONS

This study has several limitations that should be considered when interpreting its findings. First, and most significantly, the statistical comparisons in this paper are based on only five random seeds per model. Although five random seeds are sufficient to compute summary statistics such as the mean and standard deviation and to identify consistent performance trends, they provide only limited statistical power for formal hypothesis testing. This limitation is evident in the Wilcoxon signed-rank test reported in Section VI-J, where a consistent directional difference across all five seeds was still insufficient to achieve statistical significance at the conventional ($\alpha = 0.05$) threshold. Consequently, any reported improvements or degradations in performance should be interpreted as observations from a limited-sample experimental study rather than as statistically established population-level effects.

A second limitation is the scope of the evaluation itself. All experiments were conducted on a single two-dimensional incompressible flow reconstruction problem defined on a fixed spatial domain and using a specific collocation-point configuration consisting of two vertical sampling lines, as illustrated in Fig. 1. As a result, the observed relationship between reconstruction accuracy and uncertainty estimation may not generalize to other flow conditions, geometries, collocation strategies, three-dimensional problems, or time-dependent flows. Additional experiments would be required to determine whether similar trends hold in these more complex settings.

A third limitation concerns the evaluation of the pressure field. Quantitative pressure-reconstruction error metrics were not included in the supplied experimental results, making it impossible to assess pressure prediction accuracy numerically. For this reason, the discussion of the pressure field in Section VI-E is restricted to qualitative observations based on the spatial patterns shown in Fig. 11, and no numerical claims regarding pressure-reconstruction performance are made.

Finally, both uncertainty-aware architectures rely on modeling assumptions that introduce their own limitations. The uncertainty estimates produced by MC Dropout depend on the dropout configuration used during training and inference, but the relevant implementation details were not available in the supplied experimental records. Similarly, the Split-Head UQ-PINN employs a Gaussian negative log-likelihood objective, which assumes that prediction errors can be adequately represented by a Gaussian distribution. If the true error distribution exhibits substantial skewness, heavy tails, or other non-Gaussian characteristics, the resulting uncertainty estimates may not fully capture the underlying predictive uncertainty.

Fifth, this paper reports only the metrics and figures supplied in the accompanying experimental results; certain training hyperparameters (optimizer settings, learning rate schedule, exact network width and depth, dropout rate, and total wall-clock or GPU-hour cost per model) were not present in the supplied material and are therefore not reported here, which may limit exact reproducibility by readers without access to the underlying implementation.

Finally, the spatial uncertainty maps in Figs. 8 and 9 were interpreted only at the qualitative, visual level supported by the figures; more rigorous spatial statistical analysis of these uncertainty fields - for example, formal spatial autocorrelation testing - was outside the scope of the supplied experimental results and was not attempted here.

IX. CONCLUSION AND FUTURE SCOPE

This paper presented a controlled, five-seed experimental comparison of three physics-informed neural network architectures - a Standard deterministic PINN, an MC Dropout PINN, and a Split-Head heteroscedastic UQ-PINN - applied to the reconstruction of a two-dimensional incompressible flow field. The experiment demonstrated a clear and internally consistent accuracy-uncertainty trade-off: the Standard PINN achieved the lowest streamwise velocity RMSE (0.0426 ± 0.0062) and MAE (0.0238 ± 0.0031) of the three models and is the strongest choice when point-prediction accuracy alone is the objective, while the Split-Head UQ-PINN achieved the most favorable mean NLL (-0.895 ± 0.228) and the empirical 95% confidence-interval coverage closest to the nominal target ($94.74\% \pm 1.20\%$, versus $72.64\% \pm 2.97\%$ for MC Dropout), making it the stronger choice when reliable predictive uncertainty is the priority. This same experiment also produced a result that complicates a simple "Split-Head is best for uncertainty" narrative: MC Dropout achieved a lower (better) expected calibration error (0.442 ± 0.034) than the Split-Head model (0.663 ± 0.062), showing that different calibration diagnostics can favor different models and that no single uncertainty-aware architecture dominated across every uncertainty metric evaluated. The Wilcoxon signed-rank test comparing the Standard PINN and the Split-Head UQ-PINN yielded a p-value of 0.0625, which does not meet the conventional significance threshold of ($\alpha = 0.05$). This outcome highlights an important limitation of the study: although five random seeds are sufficient to reveal performance trends, they provide limited statistical power for drawing strong inferential conclusions about differences between models.

A key finding of this work is that none of the evaluated architectures was consistently superior across all evaluation criteria. The Standard PINN achieved the strongest point-prediction accuracy, the Split-Head UQ-PINN delivered confidence interval coverage closest to the nominal target, and MC Dropout exhibited more favorable calibration performance according to ECE. Consequently, the choice of model depends on the specific objectives of the application. Practitioners primarily concerned with minimizing reconstruction error may prefer a deterministic PINN, whereas applications requiring reliable uncertainty estimates or calibrated confidence intervals may benefit from uncertainty-aware alternatives. The results demonstrate that reconstruction accuracy, confidence interval coverage, and calibration quality represent distinct performance objectives that were not simultaneously optimized by any single model in the present study.

Several directions for future research emerge directly from the limitations identified in Section VIII. First, increasing the number of independent random-seed experiments would improve statistical power and enable more robust significance testing than is possible with a five-seed evaluation. Second, extending the analysis to additional flow configurations, including different Reynolds number regimes, three-dimensional geometries, and time-dependent flows, would help determine whether the observed accuracy-uncertainty trade-offs generalize beyond the specific reconstruction problem considered here. Third, future studies should incorporate quantitative pressure-field evaluation metrics alongside the velocity-based measures used in this work, enabling a more comprehensive assessment of reconstruction quality.

Another promising direction is the development of hybrid uncertainty-aware PINN architectures that combine the strengths of multiple uncertainty estimation strategies. For example, integrating a heteroscedastic variance head with MC Dropout-based stochastic inference could allow a single model to capture both aleatoric and epistemic uncertainty. Such an approach may provide improved uncertainty characterization and could potentially achieve stronger confidence interval coverage and calibration performance than either method alone.

Finally, a systematic investigation of collocation-point density and placement would provide valuable insight into the relationship between observational constraints, physics enforcement, and uncertainty estimation. Extending the analysis beyond the two-line collocation configuration used in this study could help determine whether the spatial uncertainty patterns

observed in Figs. 8 and 9 are primarily influenced by the distribution of training constraints, the underlying flow physics, or the uncertainty modeling strategy itself.

8 and 9 are influenced by proximity to collocation regions, a question this experiment's figures could not resolve on their own.

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